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Effectiveness of a generative AI-powered digital tutor integrated with a knowledge graph in anatomy education for nursing students: a randomized controlled trial

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Abstract

Background Human Anatomy and Histology & Embryology are foundational courses in nursing education but are often challenging for students due to their complex spatial structures, dense knowledge points, and high cognitive load. Recent advances in generative artificial intelligence (AI) and knowledge graph technologies provide new opportunities for enhancing medical education. This study aimed to evaluate the effectiveness of a blended teaching model integrating a generative AI-powered digital tutor with a knowledge graph in improving learning outcomes among nursing students.

Methods A three-arm parallel randomized controlled trial was conducted among 362 first-year nursing students at Cangzhou Medical College, of whom 301 completed the study. Participants were randomly assigned to an AI-Enhanced Group, a Blended Teaching Group, or a Traditional Teaching Group. All groups followed the same 16-week course structure based on Gagné's Nine Events of Instruction, differing only in the teaching tools and tutoring strategies used. Learning outcomes were evaluated using module examination scores, final comprehensive scores, knowledge graph comprehension scores, knowledge retention rates, case-based inference accuracy, and multidimensional questionnaires. Statistical analyses were conducted using one-way ANOVA with post-hoc comparisons.

Results Significant differences were observed among the three groups across multiple learning outcomes. The AI-Enhanced Group achieved higher module scores (82.5 ± 7.3) compared with the Blended Group (77.6 ± 8.1) and Traditional Teaching Group (74.1 ± 8.5) ($F = 28.64$, $p < 0.001$). Final comprehensive scores were also significantly higher in the AI-Enhanced Group (84.2 ± 6.8) ($F = 32.16$, $p < 0.001$). Knowledge graph comprehension scores showed a large effect size ($F = 103.72$, $p < 0.001$). Furthermore, the AI-Enhanced Group demonstrated higher knowledge retention rates (82.3%) and case-based inference accuracy (91.2%) than the comparison groups. Questionnaire results indicated stronger recognition of digital tutors, higher acceptance of blended learning, and improved autonomous learning ability.

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Conclusions The blended teaching model integrating a generative AI-powered digital tutor with a knowledge graph significantly improved nursing students' academic performance, knowledge retention, and clinical reasoning ability. This AI-enhanced instructional model may provide an effective approach for reforming foundational medical education and promoting structured knowledge learning.

Keywords Nursing Education, Anatomy and Histology & Embryology, Generative Artificial Intelligence, Knowledge Graph, Blended Teaching, Digital Tutor

Background

Educational challenges in anatomy and histology & embryology for nursing students

Human Anatomy and Histology & Embryology are fundamental courses in nursing education, providing essential knowledge for understanding human structure, physiological mechanisms, and clinical nursing practice [1]. However, these subjects are widely recognized as cognitively demanding for students because they involve complex spatial structures, intricate tissue morphology, and dynamic developmental processes [2, 3]. The large volume of fragmented knowledge points and the need to integrate "structure–function–clinical application" relationships further increase the learning difficulty.

Traditional lecture-based teaching approaches often rely heavily on static diagrams, physical models, and teacher-centered explanations [4]. Although these methods can convey foundational information, they frequently fail to support students in constructing coherent knowledge frameworks or visualizing spatial anatomical relationships [5]. As a result, many students experience difficulties in knowledge integration, long-term retention, and the transfer of theoretical knowledge to clinical problem-solving contexts. Improving the effectiveness of anatomy-related teaching has therefore become an important issue in nursing education.

Emerging applications of artificial intelligence in medical education

Recent advances in artificial intelligence (AI), particularly generative AI technologies, have introduced new opportunities for transforming medical education [6, 7]. AI-powered educational tools can support adaptive learning, intelligent tutoring, automated feedback, and personalized learning pathways [8, 9]. Previous studies have shown that AI-assisted learning systems can enhance students' engagement, promote self-directed learning, and improve academic performance in various disciplines [10, 11].

In parallel, knowledge graph technology has been increasingly applied in education to represent complex knowledge systems through structured networks [12, 13]. A knowledge graph generally organizes learning content as interconnected nodes and links. In the context of anatomy and histology & embryology education, these nodes may represent anatomical structures, histological features, embryological processes, or clinical concepts, whereas the links may indicate hierarchical, structural, functional, or clinical relationships among them. By making such relationships explicit, knowledge graphs can help learners move beyond isolated memorization and visualize how different concepts are connected, thereby facilitating knowledge organization, cognitive mapping, and conceptual integration [14]. In medical education, knowledge graphs have been used to support clinical reasoning training, curriculum design, and intelligent learning resource recommendation [15].

Despite these promising developments, most existing studies have examined either AI-based tutoring systems or knowledge graph–based learning tools independently. Research exploring the integrated application of generative AI tutoring and knowledge graph–supported learning environments remains limited, particularly in foundational medical courses such as anatomy and histology & embryology.

Research gap and study hypothesis

Although AI technologies are increasingly introduced into higher education, empirical evidence regarding their effectiveness in structured medical knowledge learning remains insufficient [16, 17]. Several limitations can be identified in the current literature.

First, many studies on AI in education rely on observational designs or short-term interventions, lacking rigorous experimental evidence such as randomized controlled trials [18]. Second, existing AI-assisted learning systems often focus on providing automated question answering or content generation but do not adequately address the challenge of structuring complex knowledge systems, which is a critical requirement in anatomy education. Third, few studies have investigated whether combining AI-powered digital tutoring with knowledge graph–based knowledge organization can improve higher-order learning outcomes such as knowledge retention and clinical reasoning ability.

To address these gaps, the present study proposes an AI-enhanced blended teaching model that integrates a generative AI-powered digital tutor with a knowledge graph–based learning environment, guided by Gagné's Nine Events of Instruction [19]. The effectiveness of this model was evaluated through a three-arm randomized controlled trial comparing an AI-Enhanced teaching

model, a conventional blended teaching model, and a traditional teaching approach among first-year nursing students [20].

We hypothesized that the AI-enhanced teaching model would:

1. Significantly improve students' academic performance compared with blended and traditional teaching models;
2. Enhance students' structured understanding of anatomical knowledge through knowledge graph–supported learning;
3. Improve higher-order learning outcomes, including knowledge retention and case-based clinical reasoning ability.

The findings of this study may provide empirical evidence for integrating generative AI technologies into foundational medical education and offer practical insights for the reform of anatomy-related teaching in nursing programs.

Methods

Study design and participants

This study employed a three-arm parallel randomized controlled trial to evaluate the effectiveness of an AI-enhanced teaching model integrating a generative AI–powered digital tutor with a knowledge graph in anatomy and histology & embryology education for nursing students.

Participants were recruited from six classes of first-year nursing students enrolled in the 2025 cohort at Cangzhou Medical College, China. All students were undertaking the course Human Anatomy and Histology & Embryology for the first time as part of their nursing curriculum.

The inclusion criteria were as follows: Students enrolled in the first year of the nursing program; No prior formal training in anatomy or histology & embryology; voluntary participation in the study and ability to complete the full teaching intervention and assessments; and basic computer literacy, defined as the ability to log into the online learning platform, access course materials, enter questions or responses, and complete online learning activities. This criterion did not require programming experience or prior technical expertise in artificial intelligence.

Students who withdrew from the course, transferred to other programs, or failed to complete all assessments during the study period were excluded from the final analysis.

The study protocol was reviewed and approved by the Ethics Committee of Cangzhou Medical College. All participants provided informed consent before participation.

Randomization and group allocation

Participants were randomly assigned to one of three groups: the AI-Enhanced Group, the Blended Teaching Group, or the Traditional Teaching Group.

Randomization was conducted using a computer-generated randomization sequence. To minimize potential contamination between students within the same classroom environment, group allocation was performed at the class level. Each class was assigned to one of the three teaching models according to the generated random sequence.

The allocation procedure was conducted by a researcher who was not involved in the teaching intervention or outcome evaluation.

Teaching intervention

All three groups followed the same 16-week teaching schedule with four class hours per week. The instructional process was designed according to Gagné's Nine Events of Instruction [40], including gaining attention, informing learners of objectives, stimulating recall of prior learning, presenting the content, providing learning guidance, eliciting performance, providing feedback, assessing performance, and enhancing retention and transfer.

The three groups differed primarily in the teaching tools and tutoring strategies employed.

In the AI-Enhanced Group, students used a generative AI-powered digital tutor integrated with a knowledge graph–based learning environment. In this study, the digital tutor was used as a course-support tool embedded in the learning environment. Its functions were limited to course-related learning support, including personalized preview tasks before class, real-time question answering during class, and adaptive feedback and customized review plans after class based on students' learning performance within the Human Anatomy and Histology & Embryology curriculum.

In the Blended Teaching Group, the same knowledge graph learning resources were provided, but tutoring and feedback were delivered primarily by instructors through conventional blended teaching approaches.

In the Traditional Teaching Group, teaching relied on conventional lecture-based instruction using textbooks, slides, and physical anatomical models without digital tutoring support.

A detailed comparison of the teaching intervention components across the three groups is presented in Table 1.

Outcome measures

To evaluate the effectiveness of the teaching interventions, multiple outcome indicators were assessed,

Table 1 Comparison of Teaching Intervention Programs Among the Three Groups

Intervention Links	AI-Enhanced Group	Blended Group	Traditional Teaching Group
Pre-class Preview	Digital tutor pushes knowledge graph + personalized tasks	Teacher pushes knowledge graph + unified preview tasks	Textbook reading + after-class exercise preview
In-class Teaching	Graph interactive demonstration + AI real-time Q&A	Graph demonstration + teacher Q&A	PPT lecture + model demonstration
After-class Tutoring	Digital tutor's wrong question analysis + customized review plan	Teacher's online Q&A + unified review materials	Offline Q&A + exercise book practice
Assessment and Feedback	AI-generated competency diagnosis report + dynamic task adjustment	Teacher's manual correction + phased feedback	Paper-based test correction + in-class comments

including academic performance, higher-order learning outcomes, and student learning perceptions.

Before the teaching intervention, a pre-test was administered to assess students' baseline course-related knowledge in Human Anatomy and Histology & Embryology. The pre-test score was used to evaluate baseline comparability among the three groups rather than as a post-intervention outcome.

The self-assessment of learning foundation was also collected before the teaching intervention as a baseline questionnaire measure. It was derived from the corresponding dimension of the Blended Teaching Effectiveness Questionnaire and reflected students' self-rated knowledge organization, information analysis, and independent learning abilities on a five-point Likert scale.

Academic performance indicators

Module scores Module examinations were conducted every four weeks during the semester, resulting in three module tests in total. Each test had a full score of 100 points. The average score across the three module examinations was calculated as the module score.

Final comprehensive score The final comprehensive score was calculated using a weighted evaluation system that included a theoretical examination (60%), practical laboratory assessment (30%), and process evaluation of learning activities (10%). The total score ranged from 0 to 100.

Knowledge graph comprehension score Students' understanding of the knowledge graph learning system was assessed using a Likert-scale questionnaire (1 = completely not understood, 5 = completely understood). The instrument included 20 items across five dimensions: knowledge structure recognition, graph navigation ability, concept association, application ability, and perceived learning support. The questionnaire demonstrated good internal consistency (Cronbach's $\alpha = 0.892$).

Higher-order learning outcomes

Knowledge retention rate To assess long-term knowledge retention, a follow-up test was conducted one month after the final examination. The knowledge retention rate was calculated as: Knowledge retention rate = (follow-up test score / final examination score) × 100%.

Case-based inference accuracy Ten clinical case-based questions were designed to evaluate students' ability to apply anatomical knowledge to clinical scenarios. For example, students were asked to analyze the mechanism of appendicitis pain migration based on anatomical structures. The percentage of correctly answered questions was calculated as the case-based inference accuracy.

Questionnaire evaluation

A Blended Teaching Effectiveness Questionnaire was specifically developed for this study based on the study objectives. The questionnaire consisted of 34 items measured on a five-point Likert scale. An English-language version of the questionnaire is provided as Supplementary File 1. The questionnaire used in this study has not been previously published elsewhere.

The instrument included three dimensions: Self-assessment of learning foundation (12 items) Evaluating students' abilities in knowledge organization, information analysis, and independent learning (Cronbach's $\alpha = 0.826$).

Student expectations (10 items) Assessing students' acceptance of and expectations toward digital tutors and knowledge graph–supported learning (Cronbach's $\alpha = 0.863$).

Learning habits (12 items) Evaluating behavioral patterns related to previewing, reviewing, and learning tool usage (Cronbach's $\alpha = 0.845$).

Statistical analysis

Statistical analyses were conducted using SPSS version 26.0 (IBM Corp., Armonk, NY, USA).

Continuous variables were expressed as mean ± standard deviation (SD), and categorical variables were presented as frequencies and percentages.

Prior to inferential analyses, the Shapiro–Wilk test was used to assess normality of continuous variables, and Levene's test was used to evaluate homogeneity of variances.

For variables meeting parametric assumptions, one-way analysis of variance (ANOVA) was used to compare differences among the three groups (AI-Enhanced Group, Blended Teaching Group, and Traditional Teaching Group). When overall ANOVA results were statistically

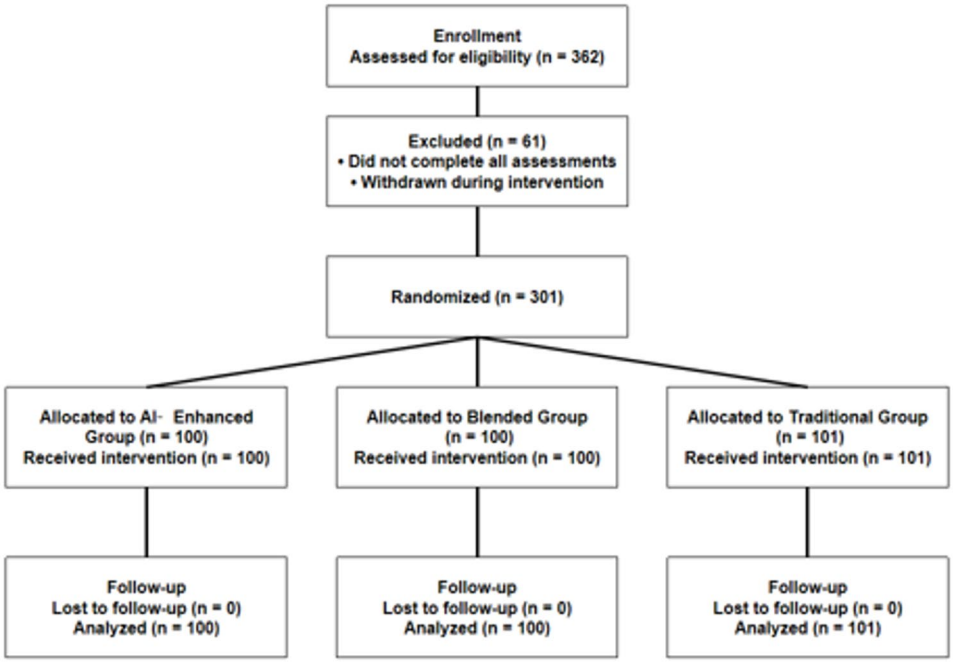


Fig. 1 CONSORT flow diagram of participant enrollment, allocation, follow-up, and analysis

significant, Bonferroni post-hoc tests were conducted for pairwise comparisons.

Effect sizes were calculated and reported as eta squared (η^2) to quantify the magnitude of group differences [21]. According to conventional benchmarks, η^2 values of 0.01, 0.06, and 0.14 were interpreted as small, medium, and large effect sizes, respectively [22].

For pairwise comparisons, mean differences and 95% confidence intervals (CIs) were reported.

Categorical variables were analyzed using the chi-square (χ^2) test.

All statistical tests were two-tailed, and p values < 0.05 were considered statistically significant.

Results

Participant flow and baseline characteristics

A total of 362 first-year nursing students were initially enrolled in the study. During the study period, students who withdrew from the course, transferred to other programs, or failed to complete all assessments were excluded. Finally, 301 participants were included in the final analysis.

The participants were randomly assigned to three groups: the AI-Enhanced Group ($n = 100$), the Blended Teaching Group ($n = 100$), and the Traditional Teaching Group ($n = 101$).

The recruitment, allocation, follow-up, and analysis of participants are illustrated in the CONSORT flow diagram (Fig. 1).

Baseline characteristics of the participants, including the pre-intervention pre-test score and self-assessment of learning foundation, are presented in Table 2. No statistically significant differences were observed among the three groups in terms of gender distribution ($\chi^2 = 0.126$, $p = 0.94$), age ($F = 0.582$, $p = 0.56$), pre-test scores ($F = 0.143$, $p = 0.87$), or self-assessment of learning foundation ($F = 0.472$, $p = 0.62$). These results indicate that the three groups were comparable at baseline.

Table 2 Comparison of Baseline Data of the Three Groups

Indicators	AI-Enhanced Group ($n = 100$)	Blended Group ($n = 100$)	Traditional Teaching Group ($n = 101$)	Test Statistic	*p* value
Gender (Male/Female, n)	9/91	10/90	9/92	$\chi^2 = 0.126$	0.94
Age (Years, x ± s)	18.2 ± 0.5	18.3 ± 0.4	18.2 ± 0.6	$F = 0.582$	0.56
Pre-test Score (Points, x ± s)	62.3 ± 8.5	61.8 ± 9.1	62.1 ± 8.8	$F = 0.143$	0.87
Self-Assessment of Learning Foundation (x ± s)	2.86 ± 0.5	2.81 ± 0.5	2.79 ± 0.5	$F = 0.472$	0.62

Academic performance outcomes

Module scores

Significant differences in module scores were observed among the three groups ($F = 28.64$, $p < 0.001$, $\eta^2 = 0.16$), indicating a large effect size.

The AI-Enhanced Group achieved the highest module scores (82.5 ± 7.3), followed by the Blended Teaching

Group (77.6 ± 8.1) and the Traditional Teaching Group (74.1 ± 8.5).

Post-hoc pairwise comparisons showed that the AI-Enhanced Group scored significantly higher than the Blended Group (mean difference = 4.9 points, 95% CI: 2.1–7.7, $p = 0.003$) and the Traditional Teaching Group (mean difference = 8.4 points, 95% CI: 5.6–11.2, $p < 0.001$). The difference between the Blended Group and the Traditional Teaching Group was not statistically significant (mean difference = 3.5 points, 95% CI: −0.4–7.3, $p = 0.09$).

Final comprehensive scores

Final comprehensive scores also differed significantly among the three groups ($F = 32.16$, $p < 0.001$, $\eta^2 = 0.18$), indicating a large effect size.

Students in the AI-Enhanced Group achieved the highest final scores (84.2 ± 6.8), followed by the Blended Teaching Group (78.9 ± 7.5) and the Traditional Teaching Group (75.2 ± 7.9).

Post-hoc comparisons indicated that the AI-Enhanced Group significantly outperformed the Blended Group (mean difference = 5.3 points, 95% CI: 2.7–7.9, $p = 0.002$) and the Traditional Teaching Group (mean difference = 9.0 points, 95% CI: 6.4–11.6, $p < 0.001$). The difference between the Blended Group and the Traditional Teaching Group was not statistically significant (mean difference = 3.7 points, 95% CI: −0.1–7.5, $p = 0.07$).

Knowledge graph comprehension

Knowledge graph comprehension scores differed significantly among the three groups ($F = 103.72$, $p < 0.001$, $\eta^2 = 0.41$), representing a very large effect size.

The AI-Enhanced Group achieved the highest comprehension score (4.54 ± 0.50), followed by the Blended Teaching Group (3.81 ± 0.70) and the Traditional Teaching Group (2.96 ± 0.88).

Pairwise comparisons showed that the AI-Enhanced Group scored significantly higher than both the Blended Group (mean difference = 0.73, 95% CI: 0.55–0.91, $p < 0.001$) and the Traditional Teaching Group (mean difference = 1.58, 95% CI: 1.36–1.80, $p < 0.001$). The Blended Group also demonstrated significantly higher scores than the Traditional Teaching Group (mean difference = 0.85, 95% CI: 0.63–1.07, $p < 0.001$).

The detailed comparison of academic outcomes among the three groups is presented in Table 3.

Higher-order learning outcomes

Knowledge retention rate

Knowledge retention rates differed significantly among the three groups ($F = 48.52$, $p < 0.001$, $\eta^2 = 0.25$), indicating a large effect size.

The AI-Enhanced Group demonstrated the highest retention rate (82.3% ± 9.5%), followed by the Blended Teaching Group (71.5% ± 10.2%) and the Traditional Teaching Group (63.8% ± 11.4%).

Post-hoc comparisons showed that the AI-Enhanced Group had significantly higher retention than the Blended Group (mean difference = 10.8%, 95% CI: 7.4–14.2, $p < 0.001$) and the Traditional Teaching Group (mean difference = 18.5%, 95% CI: 14.7–22.3, $p < 0.001$). The Blended Group also demonstrated significantly higher retention than the Traditional Teaching Group (mean difference = 7.7%, 95% CI: 4.2–11.2, $p < 0.001$).

Case-based inference accuracy

Significant differences were also found in case-based inference accuracy among the three groups ($F = 67.14$, $p < 0.001$, $\eta^2 = 0.31$), indicating a large effect size.

The AI-Enhanced Group achieved the highest accuracy (91.2% ± 7.8%), followed by the Blended Teaching Group (78.5% ± 9.3%) and the Traditional Teaching Group (67.3% ± 10.5%).

Post-hoc comparisons indicated that the AI-Enhanced Group significantly outperformed the Blended Group (mean difference = 12.7%, 95% CI: 9.4–16.0, $p < 0.001$) and the Traditional Teaching Group (mean difference = 23.9%, 95% CI: 20.2–27.6, $p < 0.001$). The Blended Group also demonstrated significantly higher accuracy than the Traditional Teaching Group (mean difference = 11.2%, 95% CI: 7.5–14.9, $p < 0.001$).

Questionnaire outcomes

The results of the blended teaching effectiveness questionnaire are summarized in Table 4.

Significant differences were observed among the three groups in several key indicators related to teaching perception and learning behavior.

Table 3 Comparison of Academic Outcomes Among the Three Groups

Outcome	AI-Enhanced ($n = 100$)	Blended ($n = 100$)	Traditional ($n = 101$)	F	p	η^2
Module score	82.5 ± 7.3	77.6 ± 8.1	74.1 ± 8.5	28.64	< 0.001	0.16
Final comprehensive score	84.2 ± 6.8	78.9 ± 7.5	75.2 ± 7.9	32.16	< 0.001	0.18
Knowledge graph comprehension	4.54 ± 0.50	3.81 ± 0.70	2.96 ± 0.88	103.72	< 0.001	0.41
Knowledge retention rate (%)	82.3 ± 9.5	71.5 ± 10.2	63.8 ± 11.4	48.52	< 0.001	0.25
Case inference accuracy (%)	91.2 ± 7.8	78.5 ± 9.3	67.3 ± 10.5	67.14	< 0.001	0.31

Values are presented as mean ± SD. η^2 indicates effect size (eta squared)

Table 4 Comparison of Core Questionnaire Indicators Among the Three Groups (x ± s, %)

Evaluation Indicators	AI-Enhanced Group	Blended Group	Traditional Teaching Group	F-value	pvalue
Knowledge Graph Comprehension	4.66 ± 0.72 (99.5%)	4.16 ± 0.80 (97.0%)	4.17 ± 0.80 (97.0%)	12.87	< 0.001
Digital Tutor Recognition	4.90 ± 0.56 (99.5%)	4.44 ± 0.62 (100.0%)	4.45 ± 0.62 (100.0%)	15.32	< 0.001
Blended Teaching Acceptance	3.68 ± 0.91 (93.1%)	3.29 ± 1.01 (81.0%)	3.31 ± 1.02 (81.2%)	6.74	0.001
Knowledge Organization Ability	2.71 ± 0.89 (43.7%)	2.42 ± 0.99 (38.0%)	2.42 ± 0.99 (37.6%)	2.86	0.06
Autonomous Learning Habits	3.34 ± 0.76 (85.1%)	2.98 ± 0.84 (74.0%)	3.00 ± 0.86 (74.3%)	7.23	0.001

The value in parentheses is the proportion of scores above 3 points (positive evaluation)

Students in the AI-Enhanced Group reported significantly higher scores in knowledge graph comprehension (4.66 ± 0.72) compared with the Blended Group (4.16 ± 0.80) and the Traditional Teaching Group (4.17 ± 0.80) (F = 12.87, $p < 0.001$).

Similarly, recognition of the digital tutor was significantly higher in the AI-Enhanced Group (4.90 ± 0.56) than in the Blended Group (4.44 ± 0.62) and the Traditional Teaching Group (4.45 ± 0.62) (F = 15.32, $p < 0.001$).

Students in the AI-Enhanced Group also demonstrated higher acceptance of the blended teaching model (3.68 ± 0.91) compared with the Blended Group (3.29 ± 1.01) and the Traditional Teaching Group (3.31 ± 1.02) (F = 6.74, $p = 0.001$).

No statistically significant difference was found among the three groups in knowledge organization ability (F = 2.86, $p = 0.06$).

However, students in the AI-Enhanced Group reported significantly stronger autonomous learning habits (3.34 ± 0.76) than those in the Blended Group (2.98 ± 0.84) and the Traditional Teaching Group (3.00 ± 0.86) (F = 7.23, $p = 0.001$).

Autonomous learning ability

Further analysis of students' self-reported learning behaviors indicated that a significantly higher proportion of students in the AI-Enhanced Group demonstrated strong autonomous learning ability.

Specifically, 89.0% of students in the AI-Enhanced Group rated their autonomous learning ability as "relatively consistent" or above, compared with 72.0% in the Blended Group and 68.3% in the Traditional Teaching Group.

The difference among the three groups was statistically significant ($\chi^2 = 23.56$, $p < 0.001$).

Between-group comparisons of the core questionnaire indicators are illustrated in Fig. 2.

Discussion

Effectiveness of the AI-enhanced teaching model

This study evaluated the effectiveness of a blended teaching model integrating a generative AI-powered digital tutor with a knowledge graph in anatomy and histology & embryology education for nursing students. The results demonstrated that the AI-enhanced teaching model significantly improved students' academic performance, knowledge graph comprehension, knowledge retention, and case-based inference ability compared with both the conventional blended teaching model and traditional teaching approaches.

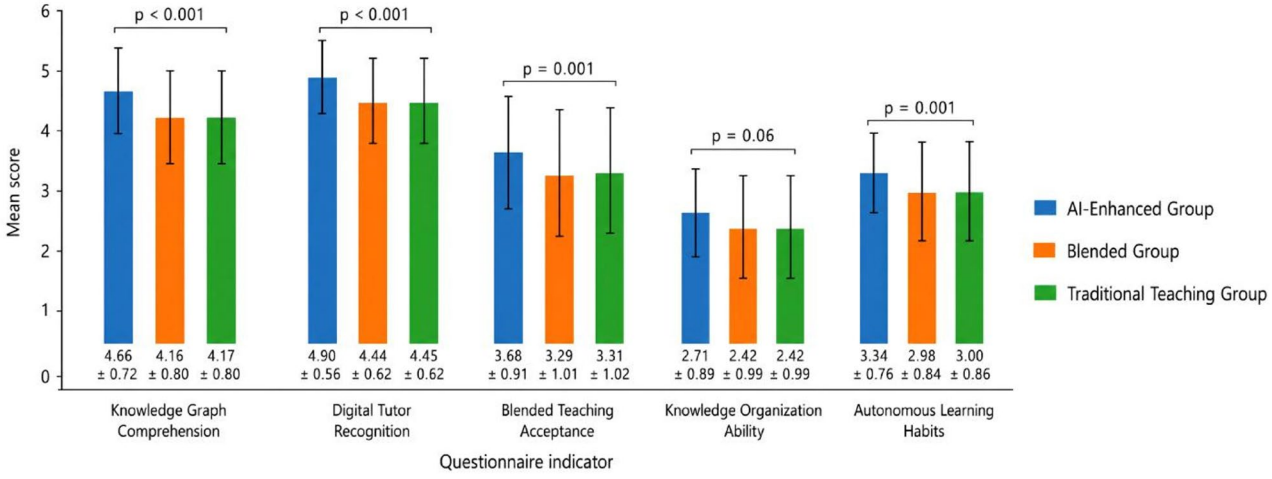


Fig. 2 Between-group comparison of core indicators from the teaching questionnaire among the three groups

In particular, the AI-Enhanced Group achieved significantly higher module scores and final comprehensive scores than the other two groups. These findings suggest that integrating AI-supported learning tools into foundational medical education can effectively enhance students' learning outcomes [6, 10]. Previous research has also indicated that AI-assisted educational systems can improve student engagement, provide adaptive learning support, and promote deeper understanding of complex concepts [8, 9, 11]. Our findings extend this evidence by demonstrating the effectiveness of an integrated AI-tutoring and knowledge-graph framework in a randomized controlled trial setting.

Importantly, the effect sizes observed in this study indicate that the intervention produced meaningful educational benefits [21]. The large effect sizes observed for several learning indicators suggest that AI-enhanced teaching may substantially influence students' learning processes and outcomes rather than merely producing marginal improvements [22].

Improvement in structured knowledge learning

One notable finding of this study is the significant improvement in knowledge graph comprehension among students in the AI-Enhanced Group. Anatomy and histology & embryology involve highly structured knowledge systems with complex hierarchical relationships among concepts [2, 3]. Traditional teaching methods often present these concepts in fragmented ways, which may hinder students from constructing coherent cognitive frameworks [4, 5].

Knowledge graphs provide a structured visual representation of relationships among concepts, enabling students to organize knowledge more effectively [12, 14]. Previous studies have suggested that knowledge graph–based learning environments can support cognitive mapping and knowledge integration [23, 24]. The findings of this study further support this perspective by showing that students who used knowledge graph-supported learning tools demonstrated significantly higher comprehension scores than those receiving conventional instruction.

Moreover, the integration of knowledge graphs with AI tutoring appears to enhance students' ability to navigate and utilize structured knowledge systems. Through personalized learning guidance and interactive exploration of knowledge nodes, students may develop a more coherent understanding of anatomical structures and their relationships. This structured learning approach may explain the large effect size observed for knowledge graph comprehension in the present study.

Enhancement of higher-order learning outcomes

Beyond improvements in academic scores, the AI-enhanced teaching model also demonstrated clear advantages in higher-order learning outcomes. Students in the AI-Enhanced Group achieved significantly higher knowledge retention rates and case-based inference accuracy than those in the other groups.

Knowledge retention is a critical indicator of meaningful learning in medical education [25, 26]. The higher retention rate observed in the AI-Enhanced Group suggests that AI-supported learning environments may facilitate deeper cognitive processing and more durable memory formation [27]. By providing continuous feedback, personalized review recommendations, and adaptive practice tasks, AI digital tutors can help students identify knowledge gaps and reinforce key concepts during the learning process [28, 29].

Similarly, the significant improvement in case-based inference accuracy indicates enhanced clinical reasoning ability [30, 31]. Anatomy education should not only focus on memorization of structures but also emphasize the application of anatomical knowledge to clinical situations [32]. The integration of AI-generated clinical scenarios and knowledge graph navigation may help students connect theoretical knowledge with clinical reasoning processes, thereby improving their ability to analyze and solve clinical problems.

Mechanisms of AI-supported learning in medical education

The effectiveness of the AI-enhanced teaching model observed in this study may be explained by several educational mechanisms.

First, AI digital tutors provide real-time feedback and adaptive learning support, which are difficult to achieve in traditional classroom settings [28, 33]. Immediate feedback has been widely recognized as an important factor in promoting effective learning, as it allows students to correct misconceptions and reinforce correct understanding during the learning process [28, 34].

Second, the AI tutoring system enables personalized learning pathways by analyzing students' learning behavior and performance data [29, 35]. Personalized learning has been shown to increase learning efficiency and motivation, particularly in complex subjects such as anatomy [35, 36].

Third, the integration of AI tutoring with knowledge graphs supports structured knowledge construction [14, 23]. By linking individual knowledge points within a coherent conceptual network, students are better able to understand relationships among concepts and build a hierarchical knowledge framework.

Finally, AI-supported learning environments may promote self-directed learning behaviors [35, 37]. The

questionnaire results in this study indicated that students in the AI-Enhanced Group reported higher levels of autonomous learning ability, suggesting that AI tools may encourage students to actively engage in the learning process.

Educational implications

The findings of this study have several implications for medical education. First, integrating generative AI technologies into foundational medical courses may provide an effective strategy for addressing long-standing challenges in anatomy teaching, such as high cognitive load and fragmented knowledge presentation [6, 7, 38].

Second, the results highlight the importance of combining AI technologies with structured knowledge representation tools, such as knowledge graphs [12, 13, 14]. Rather than using AI solely for automated question answering or content generation, integrating AI with structured knowledge systems may produce more meaningful educational benefits.

Third, the AI-enhanced teaching model evaluated in this study may provide a scalable approach for improving teaching quality in large classes. AI digital tutors can offer individualized learning support to large numbers of students simultaneously, thereby alleviating the limitations of traditional teacher-centered instruction.

These findings suggest that AI-supported blended teaching models may play an important role in the future development of medical education [19, 39].

Limitations and future research

Despite the promising findings, several limitations should be acknowledged.

First, the study was conducted at a single medical college, which may limit the generalizability of the results to other institutions or educational contexts. Future multicenter studies are needed to validate the effectiveness of the proposed teaching model across different educational settings.

Second, the intervention was implemented over a single semester, and long-term learning outcomes beyond the follow-up test were not evaluated. Longitudinal studies may provide further insights into the sustained impact of AI-enhanced teaching on students' clinical competence [40].

Third, although AI tools provide valuable learning support, excessive reliance on AI technologies may potentially reduce students' independent problem-solving abilities [41, 42]. Future research should explore strategies for balancing AI-supported learning with the development of critical thinking skills.

Fourth, this study evaluated the educational application of the AI-enhanced teaching model but did not independently validate the technical development process of the generative AI-powered digital tutor or systematically audit the factual accuracy of each AI-generated response. In addition, the study did not separately examine how differences in students' prompting behaviors may have influenced learning outcomes. Future studies should describe the development, content validation, accuracy-monitoring procedures, and student training protocols for AI tutors in greater detail.

Conclusions

The blended teaching model integrating Generative AI-powered digital tutors with knowledge graphs significantly enhances nursing students' academic performance, structured knowledge cognition, autonomous learning ability, and clinical reasoning. Given its demonstrated effectiveness, this model is recommended for broader implementation in basic medical education and for expansion into related courses. Continued refinement of AI functionalities and graph-based content is essential to support sustainable improvement in nursing education.

Abbreviations

AI	Artificial intelligence
ANOVA	Analysis of variance
SD	Standard deviation
η^2	eta squared
χ^2	chi-square

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

CZ Conceptualization, Methodology, Writing – Original Draft. .**JZ** Data Curation.**JL** Formal Analysis.**WZ** Investigation.**YP** Writing – Review & Editing, Supervision.All authors read and approved the final manuscript.

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Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study protocol was approved by the Ethics Committee of Cangzhou Medical College (Approval No. CZMC-2025-06-20). All procedures involving human participants were conducted in accordance with the ethical standards of the institutional research committee and the principles of the Declaration of Helsinki. Written informed consent was obtained from all participants prior to their participation in the study.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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